UNIVERSITÀ DI PISA

Dipartimento di ingegneria dell'informazione

Master's degree in Artificial Intelligence and data engineering

Smart maintenance for automotive businesses

Corso: Business and Project Management
Ambito: AI-Driven Business Venture

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Gitub Repository: `https://github.com/marcolari06-maker/BPM-project`

Abstract

In this short paper we present how we intend to organize an AI driven startup in the automotive sector, defining the problem we mean to address and the solutions provided. Furthermore we provide a competitor analysis, a business model canvas, a market estimation and a negative backstacking; evaluating the quality of our own project model.

Contents

1 Introduction to the Problem

1.1 Problem and drivers

The problem addressed in this paper refers to the inefficiencies and consequent high costs of vehicle maintenance in automotive companies, such as transportation, logistic and rental businesses.

Problems

- High maintenance costs; costs rose by 8.6% year over year in 2025, reaching an industry average of 0.215$ per mile for commercial vehicles in USA. (ATRI report)
- Downtime and overhead of maintenance work, the average mileage traveled between brakdowns fell to 36.500 from 38.500, indicating that the matter is also worsening over time. (ATRI report)
- Parts reperability issues. The rising tariffs on importation and the supply chain directly impact the availability of spare parts. (ATRI report)

Drivers

- Unexpected failures of critical components (engine, battery pack, transmission). Companies can't predict failures and are forced to wait for them to happen before taking action.
- Unnecessary repairs or replacements to maintain security standard due to lack of predictive maintenance, which leads to periodic schedules or over-maintenance.

1.2 Solution and innovations

Given the above problems and drivers, we aim to provide a solution based on a machine learning model that enables to treat such matters with statistical evidence, such that overhead is reduced due to prediction of failures and costs are reduced with reduction on unnecessary maintenance work. A machine learning approach is suitable for this kind of problem because unlike fixed rules, heuristic or treshold based approaches, it can learn patterns from data to adapt to situations without the need to make rules for every possible scenario. Furthermore it makes the system much more adaptive and scalable, since it can be continuously improved with the collected data, and the data income comes with the normal operation of the vechicles.

This kind of approach is innovative in small automotive companies, since is usually adopted as an internal solution by larger automotive companies, such as OEMs and Tier 1 suppliers. Instead we aim to provide a B2B service that can be adopted by smaller realities, such as logistic companies, rental companies and small fleet owners.

- Small companies don't have the resources to develop such a system internally.
- Previous solutions are heavy on the budget and could be threatening for a upcoming business.

Instead in the proposed solution:

- The system is provided as a service, with a subscription model, such that the company can pay for the service without having to develop it internally.
- The system is designed to be lightweight and easy to integrate with existing systems, such that it can be adopted without major changes to the existing infrastructure.

- Even with the cost of the service the overall cost of maintenance is expected to be lower than the current cost, due to the reduction of downtime and unnecessary maintenance work.

1.3 Assumptions

- Companies agree to install sensors, or already have them andshare data with the service provider.
- Operational costs of the service are lower than the costs of maintenance even for a small number of vehicles, remaining in line with the target of small companies.
- Fleet managers hold sufficient operational budget authority to adopt a new SaaS tool without requiring lengthy corporate procurement approval — relevant because our target includes small-to-mid fleets (30–200 vehicles).
- Enough data is available to train the machine learning model before the service is launched, using customer data only for prediction and future improving of the model.
- The system isn't affected by high risk regulations such as the EU AI Act, since it doesn't interfere directly with the driving capability of a vehicle. This moderate costs

2 Competitors and Positioning

2.1 Competitive landscape mapping

We identify three distinct competitive tiers relevant to a predictive maintenance venture targeting small-to-mid commercial fleets.

1. **Direct Competitors:** Startups and scale-ups offering AI-driven telematics solutions for predictive maintenance, such as:
    - **Carly enterprise:** An affirmed company in the automotive telematics space. Provides predictive maintenance as well as cost calculation, all made for fleets.
    - **Steer AI:** A startup that provides predictive maintenance for fleets, with a focus on electric vehicles.
2. **Incumbent / OEM:** Established automotive manufacturers and Tier 1 suppliers that offer proprietary predictive maintenance solutions, such as:
    - **Bosch Predictive diagnostics::** a proprietary software solution that continuously monitors component and system health using onboard and cloud data, aimed at preventing breakdowns before they occur
    - Similar proprietary offerings exist from other Tier-1 suppliers and OEMs.
3. **Non AI Substitute Solutions:** Traditional maintenance methods, including:
    - Fixed-interval preventive maintenance, still used by the large majority of small and mid-sized fleets. 35% higher costs and 40% more breakdowns annually compared to fleets on optimized programs.
    - Manual visual inspection, costing an estimated \$100–\$300 per vehicle per inspection cycle and administrative overhead
    - Reactive maintenance (repair only after failure), the most expensive approach; it can cost 3 to 9 times more than the same service performed proactively

2.2 Assessing Competitors

The competitors are evaluated according to their value proposition, target customers, technological scope, integration model and potential limitations.

2.2.1 Carly Enterprise: Value Proposition, Strengths and Weaknesses

Carly Enterprise's main value proposition is to make vehicle diagnostics and maintenance-related information accessible to professional users through a mobile application, dashboard or API. The platform aims to transform technical vehicle data into actionable information, such as fault severity, probable root cause, repair priority and estimated cost.

However, Carly Enterprise also presents potential weaknesses from the perspective of the target market considered in this project.

2.2.2 SteerAI: Value Proposition, Strengths and Weaknesses

SteerAI's value proposition is based on the integration of artificial intelligence into multiple automotive processes. In the fleet context, the platform aims to combine vehicle health monitoring, predictive maintenance and operational workflows in a unified environment.

The main limitation of this approach is the breadth of the platform.

2.3 Competitor Comparison

Table 1 summarizes the main differences between the selected competitors and the proposed venture.

Table 1: Comparison of competitors and the proposed venture

Competitor	Type	Main value proposition	Main strength	Potential weakness
Carly Enterprise	Direct AI competitor	Vehicle diagnostics, fault prioritization and predictive maintenance	Large diagnostic-data base and API/mobile access	May focus primarily on diagnostic codes and may offer more functionality than small fleets require
SteerAI	Direct AI competitor	AI platform for diagnostics, fleet operations and automotive workflows	Broad integration across automotive activities	Potential complexity and limited publicly available evidence on predictive performance
Bosch Predictive Diagnostics	OEM/Tier-1 incumbent	Continuous monitoring of vehicle and component health	Technical expertise and OEM relationships	Possible ecosystem dependence and lower accessibility for independent fleets
Fixed-interval maintenance	Non-AI substitute	Simple and familiar scheduled maintenance	Low technological complexity	Does not adapt to the actual condition of individual vehicles
Reactive maintenance	Non-AI substitute	Repair only when a failure occurs	No investment in predictive systems	High exposure to downtime, emergency repair and replacement costs
Proposed venture	AI-driven SaaS	Accessible predictive maintenance for small and medium-sized multi-brand fleets	Focused product, lightweight integration and hardware-agnostic positioning	No proprietary large-scale dataset and predictive performance still to be validated

2.4 Strategic Positioning

The proposed venture should not position itself as more accurate than Carly Enterprise, SteerAI or Bosch before obtaining evidence from a real fleet pilot. These competitors have access to larger datasets, established partnerships or broader commercial resources.

Instead, the proposed differentiation is based on two strategic dimensions. First, the service is designed specifically for small and medium-sized multi-brand fleets that may not have the resources to implement a complex enterprise platform. Second, the system is intended to be lightweight and hardware-agnostic, allowing integration with existing OBD or telematics infrastructure wherever possible.

The venture therefore adopts a focused differentiation strategy rather than a cost-leadership strategy. Its competitive promise is not to offer the most complete automotive platform, but to make predictive maintenance easier to adopt and economically accessible for smaller operators.

This positioning must be validated through interviews and pilot projects. The venture can be considered competitively promising only if fleet managers confirm that the main obstacles of existing solutions are excessive complexity, integration costs, lack of multi-brand compatibility or insufficient affordability.

2.5 Validating competitiveness

To demonstrate realistic competitive viability, we provide three arguments: a quantitative cost-benefit comparison, an explanation of how we compensate for data-scale disadvantages, and an empirical validation plan.

Cost-benefit comparison

Using ATRI 2026 industry benchmarks, repair and maintenance costs average $0.215 per mile, approximately $17,200 per vehicle per year at 80,000 miles . Unplanned breakdowns occur on average every 36,891 miles, with reactive repairs cost 3-9 times more than proactive service. A subscription priced at EUR 8–15/vehicle/month ($108-$192/year) would be economically viable even with conservative savings of 5-10% on R&M and 10-20% on downtime, yielding a net benefit of approximately $770-$2,500 per vehicle per year (5-15 times return on subscription).

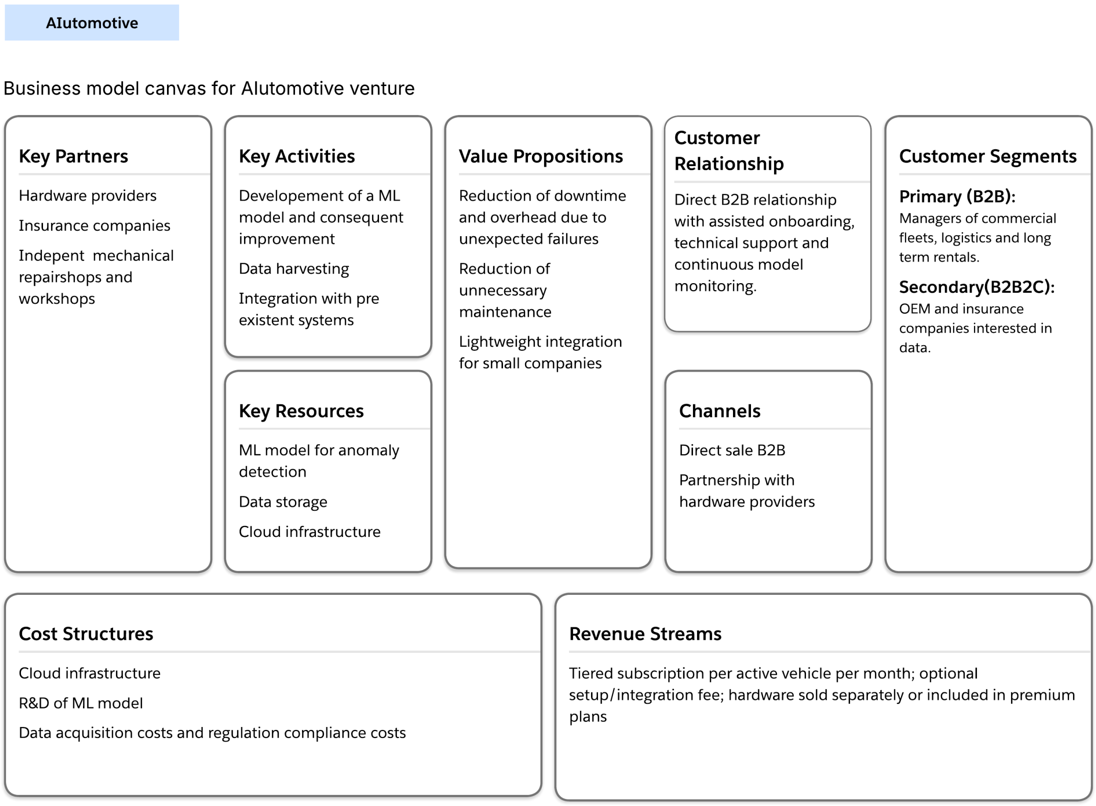
Compensating for data-scale disadvantage

Despite lacking the proprietary data volume of Carly, SteerAI, or Bosch, we mitigate this through: (i) pretraining on open predictive-maintenance datasets (e.g. NASA) before fine-tuning on real fleet data; (ii) hardware-agnostic, multi-brand integration that serves mixed fleets underserved by OEM-locked solutions; (iii) emphasis on interpretable, actionable warnings over black-box accuracy; and (iv) partnerships with telematics providers or fleet associations.

Empirical validation plan

Competitiveness claims will be tested through: (a) 15-30 customer discovery interviews to confirm that existing solutions are too complex, expensive, or brand-restricted for small fleets (success: > $60% confirm at least one obstacle); (b) a 3-6 month pilot with 50-100 vehicles to measure actual reduction in breakdowns and R&M costs (success: statistically significant improvement, p < 0.1$); and (c) optional A/B comparison between predictive and fixed-schedule maintenance on vehicle subsets. Until validated, competitiveness remains a hypothesis supported by economic plausibility and strategic differentiation, not an established fact.

3 Business model canvas



3.1 AI Assets and Economic Logic

The AI component creates value through the relationship between data, models and computational infrastructure. More reliable vehicle data can improve anomaly detection; better predictions can increase customer value; higher customer adoption can generate more data for further model improvement. This creates a reinforcing learning mechanism, although it should not be assumed automatically.

The business model is scalable if the platform can serve additional vehicles without a proportional increase in personnel, integration effort or infrastructure costs. A standardised software architecture, reusable integrations and automated model monitoring are therefore important. However, hardware installation, customer-specific configurations and fragmented vehicle data may limit scalability.

The venture should measure unit economics through indicators such as customer acquisition cost, monthly recurring revenue per vehicle, gross margin per vehicle, churn rate and customer lifetime value. Feasibility improves when the lifetime value of a customer clearly exceeds acquisition and onboarding costs, while the gross margin remains sufficient to finance model development and support.

3.2 Machine Learning Pipeline

3.2.1 Overview

The predictive maintenance system relies on two distinct pipelines: a **training pipeline**, executed offline to build and validate the model, and an **inference pipeline**, executed in production to generate real-time predictions on incoming telemetry data. Separating these two pipelines is essential both for engineering clarity and to prevent inconsistencies between how features are computed during training and during deployment.

3.2.2 Training Pipeline

1. **Data Ingestion:** raw telemetry (OBD-II codes, sensor readings such as engine temperature, vibration, battery voltage) and maintenance logs are collected per vehicle, timestamped, and stored in a structured format (vehicle ID, timestamp, sensor readings, maintenance events).

2. **Data Cleaning:** removal of duplicate readings, handling of missing sensor values (via forward-fill within a vehicle's own history, never via global imputation that could leak information across vehicles), and outlier filtering based on physically plausible sensor ranges.

3. **Labeling:** for each vehicle, a target variable is defined, either as *Remaining Useful Life* (RUL, regression) or as a *binary failure-within-horizon* flag (classification, e.g. "will this component fail within the next 30 days?"). Labels are constructed strictly using each vehicle's own historical failure records.

4. **Feature Engineering:** rolling statistics (mean, standard deviation, trend slope) are computed over fixed historical windows (e.g. last 7/30 days) using only past observations relative to each timestamp.

5. **Train/Validation/Test Split:** performed with strict temporal and group-based separation (see Section 3.2.4 for details).

6. **Model Training:** candidate models (gradient boosting for tabular features, or LSTM/Transformer architectures for raw sequential data) are trained on the training set only.

7. **Hyperparameter Tuning:** performed via cross-validation restricted to the training set, using time-aware or group-aware folds, never on the held-out test set.

8. **Evaluation:** final performance is measured once on the untouched test set, using metrics appropriate to the task (RMSE/MAE for RUL regression; precision, recall and F1 for failure classification, given the expected class imbalance).

3.2.3 Inference Pipeline

1. **Real-Time Data Collection:** incoming telemetry from a vehicle is streamed or batched at regular intervals.

2. **Feature Computation:** the same feature transformations defined during training (identical rolling windows, identical scaling parameters fitted on the training set) are applied to the incoming data, ensuring train/serve consistency.

3. **Prediction:** the trained model outputs either an estimated RUL or a failure probability within the defined horizon.

4. **Alert Generation:** predictions are filtered through a decision threshold (calibrated during validation, not on test data) and translated into prioritized, human-readable alerts for the fleet manager.

5. **Feedback Loop:** confirmed outcomes (actual failures, false alarms, maintenance actions taken) are logged and periodically fed back into the training pipeline for model retraining, always respecting the same leakage-prevention rules applied to the original training data.

3.2.4 Data Leakage Prevention

Data leakage is a critical risk in predictive maintenance settings, where naive splitting strategies can produce artificially high validation performance that does not generalize to real deployment. The following safeguards are applied throughout the pipeline.

- **Group-based splitting (by vehicle):** data from the same vehicle must never appear in both the training and test sets, even at different timestamps. Splitting must be performed at the vehicle level (e.g. 70% of vehicles for training, 15% for validation, 15% for test), not at the observation level. Random row-level splitting would leak vehicle-specific patterns (driving style, component batch, maintenance history) between sets, inflating apparent accuracy.

- **Temporal splitting:** within the training vehicles, features must be computed using only information available up to the prediction timestamp. Rolling windows, cumulative statistics or trend features must never include future observations relative to the point being predicted.

- **No global normalization:** scaling and normalization parameters (mean, standard deviation, min/max) must be fitted exclusively on the training set and then applied unchanged to validation and test sets. Fitting a scaler on the full dataset before splitting is a common and subtle leakage source.

- **Target leakage avoidance:** features that are only recorded at or after the failure event (e.g. a diagnostic code generated as a consequence of the failure itself) must be excluded from the feature set, since they would not be available at prediction time in a real deployment scenario.

- **Cross-validation strategy:** hyperparameter tuning must use group-aware and time-aware cross-validation (e.g. `GroupKFold` or a rolling-origin time series split), rather than standard k-fold, to preserve the same separation guarantees as the main train/test split.

- **Consistent feature computation at inference time:** any transformation applied to raw data during training (imputation rules, rolling window definitions, encoding of categorical variables) must be implemented identically in the inference pipeline, ideally through a shared, versioned feature-engineering module, to avoid training/serving skew.

These safeguards are particularly relevant given the assumption, introduced in Section 1, that the model may initially rely on open datasets (e.g. NASA Turbofan) before sufficient proprietary fleet data is collected. Since such datasets differ in domain from commercial road vehicles, any transfer learning or fine-tuning step must preserve the same group- and time-based separation logic when the model is later retrained on real customer data, to avoid conflating benchmark validation performance with production-readiness.

3.3 Critical Assumptions and Interdependencies

The business model depends on several critical assumptions:

- Vehicle data is available, sufficiently accurate and legally usable.

- The model can detect relevant anomalies with enough precision to justify customer adoption.

- Fleet managers are willing to change from reactive or scheduled maintenance to data-supported decisions.

- Integrations with existing systems can be completed without excessive time or cost.
- Customers perceive the reduction in downtime and unnecessary maintenance as greater than the subscription and implementation costs.
- Hardware providers, repair shops, OEMs and insurers are willing to cooperate under commercially acceptable conditions.

These assumptions are interdependent. For example, model accuracy depends on data quality, while data quality depends on hardware reliability and customer usage. Customer willingness to pay depends on measurable economic benefits, but those benefits can only be demonstrated through successful integration and sufficient operating data. Similarly, scalability depends on standardising deployments, while standardisation may be difficult if vehicles and customer systems differ substantially.

3.4 Feasibility and Scalability Assessment

The model is potentially feasible because it addresses a costly and recurring operational problem and generates recurring revenue. Its strongest economic logic is the link between preventive predictions and reduced downtime. However, feasibility should be tested through pilots rather than assumed: initial pilots should compare a customer's maintenance performance before and after implementation using indicators such as unexpected failures, downtime hours, maintenance expenditure and vehicle utilisation.

Scalability is strongest on the software side. Once the platform and model are operational, adding vehicles can increase recurring revenue without requiring a complete redesign of the service. The main scalability constraints are data heterogeneity, hardware deployment, customer-specific integration and the need to maintain prediction quality across different vehicle fleets.

The venture should therefore follow a staged growth path. It should first focus on a limited group of fleet customers and a defined set of vehicle types. After validating the value proposition and unit economics, it can expand through standardised integrations, hardware partnerships and indirect distribution via OEMs or insurers. This approach reduces execution risk and ensures that growth is based on demonstrated operational and financial performance.

4 Addressable market analysis

This section quantifies the economic opportunity for the proposed venture by estimating the Total Addressable Market (TAM), Serviceable Addressable Market (SAM), and Serviceable Obtainable Market (SOM) adopting a bottom-up methodology. We started from the number of potential customer fleets in the United States then multiplied by the expected annual subscription revenue per vehicle, consistently with the customer segment and pricing assumptions defined in Sections 1–3

4.1 Data source and scope

The primary data source for fleet counts is the FMCSA Company Census (dataset az4n-8mr2), which reports active registered motor carriers by fleet size updated on August 2026. According to this registry, there are 2,230,000 active carriers in the U.S., of which 700,000 are interstate carriers and 330,000 hold active common (for-hire) operating authority. Fleet size distribution shows that 40,000 carriers report 21–100 power units and 8,000 carriers report 101+ power units. The analysis focuses on for-hire carriers with 30–200 vehicles, as this segment matches our target (small-to-mid fleets that lack resources to develop predictive maintenance internally but have sufficient operational budget to adopt a SaaS tool without long procurement approval). Carriers with fewer than 30 vehicles are excluded because they typically operate with informal maintenance practices and limited telematics adoption. Carriers with more than 200 vehicles are excluded because they often have proprietary solutions from OEMs or Tier-1 suppliers (e.g., Bosch Predictive Diagnostics) or internal data science teams, making them less suitable for our proposal.

4.2 Total Addressable Market (TAM) Estimation

To estimate the number of carriers in the 30–200 vehicle range, we apply conservative proportional assumptions to the FMCSA buckets: • 21–100 bucket (40,000 carriers): We assume 55% of these carriers fall in the 30–100 range (the remainder are 21–29 vehicles). This yields approximately 22,000 carriers. • 101+ bucket (8,000 carriers): We assume 35% of these carriers fall in the 101–200 range (the remainder are 201+ vehicles, which are enterprise fleets less relevant to our positioning). This yields approximately 2,800 carriers. Summing these gives a TAM of approximately 24,800 carriers in the U.S. with 30–200 vehicles. To convert carriers to vehicles, we assume an average fleet size of 65 vehicles (midpoint of the 30–200 range, slightly conservative to account for the right-skewed distribution where more carriers cluster near the lower end). This yields an estimated 1,612,000 vehicles in the TAM segment. Applying the midpoint subscription price of USD 12.5/vehicle/month, the annual revenue per vehicle is USD 150. Therefore:

$$\text{TAM} = \underbrace{1{,}612{,}000}_{\text{vehicles}} \times \underbrace{150}_{\text{USD/vehicle/year}} \approx 241.8 \text{ million USD/year} \tag{1}$$

4.3 Serviceable Addressable Market (SAM) Estimation

Not all TAM carriers are immediately addressable. Based on the assumptions in Section 1.3 and 3.2, we exclude carriers that: • Operate primarily intrastate with no interstate authority (less likely to invest in advanced telematics). • Are not multi-brand (single-brand fleets may prefer OEM-locked solutions like Bosch) . • Lack existing telematics or OBD-II data access (our solution assumes sensor data is already available or can be installed with minimal overhead) . We estimate that 40

$$\text{SAM} = \underbrace{1{,}612{,}000}_{\text{vehicles}} \times \underbrace{0.40}_{\text{proportion}} \approx 645{,}000 \text{ vehicles} \tag{2}$$

$$\text{SAM (USD/year)} = 241.8 \text{ million} \times 0.40 \approx 96.7 \text{ million USD/year} \tag{3}$$

4.4 Serviceable Obtainable Market (SOM) Estimation

Given the early-stage nature of the venture (no proprietary large-scale dataset, predictive performance still to be validated via pilots as stated in Section 2.5), a realistic market penetration rate by Year 3 is 3% of the SAM. This is consistent with typical SaaS adoption curves for niche B2B verticals and aligns with the staged growth path described in Section 3.3 (start with a limited group of pilot customers, then expand through standardized integrations).

$$\text{SOM (vehicles, Year 3)} = 645{,}000 \times 0.03 \approx 19{,}350 \text{ vehicles} \tag{4}$$

$$\text{SOM (USD/year, Year 3)} = 96.7 \text{ million} \times 0.03 \approx 2.9 \text{ million USD/year} \tag{5}$$

4.5 Investment justification

The SOM of 2.9 million USD/year by Year 3 represents a credible revenue base for an early-stage AI venture. Using ATRI benchmarks, repair and maintenance costs average USD 0.215/mile, or approximately USD 17,200/vehicle/year at 80,000 miles. A subscription priced at USD

108–192/vehicle/year would be economically viable even with conservative savings of 5–10% on R&M and 10–20% on downtime, yielding a net benefit of approximately USD 770–2,500/vehicle/year (5–15× return on subscription cost), as assumed in the previous section. (da riguardare tutta questa parte? potremmo toglierla). This implies that the customer lifetime value (LTV) significantly exceeds the customer acquisition cost (CAC) and onboarding expenses, satisfying the feasibility condition stated in Section 3.3 (LTV » CAC, gross margin sufficient to finance model development). The market opportunity therefore justifies the investment and development effort required to launch the venture, provided that the pilot validation plan (15–30 customer discovery interviews, 3–6 month pilot with 50–100 vehicles) successfully confirms the assumed pain points and cost savings .

Level	Carriers	Vehicles	Annual Revenue (USD)	Logic
TAM	∼24,700	∼1,605,000	∼240.8 M	All U.S. for-hire carriers with 30–200 vehicles; USD 150/vehicle/year
SAM	∼9,900	∼642,000	∼96.3 M	40% of TAM: multi-brand, digital-ready, SaaS budget authority
SOM (Y3)	∼300	∼19,300	∼2.9 M	3% of SAM by Year 3; staged growth via pilots and standardized integrations

Table 2: Market sizing summary: TAM, SAM, and SOM.

5 Personas and Validation Strategies

This section defines the user personas that will guide market testing and describes the validation strategy for the proposed venture. The analysis still focuses on small-to-mid commercial fleets (30–200 vehicles) in the United States and treats all claims about customer adoption, data availability and economic benefits as hypotheses to be tested.

5.1 Validation objectives

The primary objectives of the market test are to:

- Confirm that unplanned downtime and reactive maintenance are perceived as high-priority, costly problems by the target segment.
- Validate that existing alternatives (fixed-interval maintenance, OEM solutions, enterprise telematics platforms) are considered too complex, expensive or brand-restricted for small-to-mid fleets.
- Assess whether target fleets have or can provide sufficiently reliable OBD or telematics data with acceptable integration effort.
- Estimate willingness to pay within the proposed range of USD 8–15 per vehicle per month (approximately USD 96–180 per year).
- Obtain early evidence that a pilot deployment can generate measurable improvements in downtime, emergency repairs or total repair-and-maintenance expenditure.

5.2 User Personas

Two primary personas are defined to guide customer discovery, MVP design and pilot recruitment. Both personas are fictitious but representative of real decision-makers and users in the target segment.

5.2.1 Persona 1 - Fleet manager

Name and profile: Peter Parker, Fleet Manager at a regional logistics company operating 80 commercial vehicles. **Role:** Responsible for fleet availability, maintenance budget, on-time delivery performance and selection of operational tools. Approves or co-approves software expenditures for fleet management and maintenance. **Context:** Manages a mixed fleet of 30–200 vehicles. Receives data from telematics providers, workshops and drivers, but often plans maintenance using fixed schedules and fragmented spreadsheets or legacy software. **Goals:**

- Minimize unplanned downtime and breakdown-related delivery delays.
- Make maintenance costs more predictable and reduce emergency repairs.
- Maintain high vehicle utilization without excessive over-maintenance.
- Justify technology investments to senior management with clear ROI.

Frustrations:

- Unexpected component failures (engine, transmission, battery) that disrupt operations.
- High and volatile repair costs, especially for urgent interventions.
- Lack of integrated, actionable insights from existing telematics or maintenance tools.
- Enterprise platforms perceived as too expensive, complex or tied to specific OEMs.

Job-to-be-done: "When I manage a mixed commercial fleet, I want to identify vehicles likely to fail before a breakdown occurs, so that I can schedule maintenance with minimal disruption and cost." **Decision power:** Can initiate a pilot or approve a low-cost SaaS subscription independently; larger investments require management or procurement approval. This aligns with the assumption in Section 1.3 that fleet managers in small-to-mid fleets hold sufficient operational budget authority to adopt a new SaaS tool without long corporate procurement processes.

5.2.2 Persona 2 - Maintenance manager/Workshop manager

Name and Profile: Steve Rogers, Maintenance Manager at a rental and logistics fleet with 120 vehicles. **Role:** Plans and coordinates all maintenance activities, manages internal technicians and external workshops, verifies anomalies, records repairs and downtime. **Context:** Handles competing priorities between scheduled maintenance and urgent repairs. Uses OBD data, GPS/telematics platforms, spreadsheets or dedicated fleet-maintenance software.

Goals:

- Detect potential failures early enough to schedule interventions proactively.
- Reduce the number of emergency repairs and associated overtime costs.
- Organize spare parts and workshop workload more efficiently.

- Avoid false alarms that waste time and undermine trust in the system.

Frustrations:

- Alerts that are difficult to interpret or not linked to clear actions.
- Fragmented or low-quality data from multiple sources.
- Long and costly integration projects with existing systems.
- Complex dashboards that do not fit daily operational routines.

Job-to-be-done: “When the system detects a possible component failure, I want an understandable and actionable warning, so that I can inspect or service the right vehicle before it becomes unavailable.

Adoption condition: Will actively use the system only if alerts are interpretable, prioritized and connected to concrete maintenance actions. This reflects the critical assumption in Section 3.2 that customers must perceive the reduction in downtime and unnecessary maintenance as greater than the subscription and implementation costs .

5.3 Key hypotheses to validate

The market test focuses on the following hypotheses, which are directly derived from the problem definition, competitive positioning and business model described in Sections 1–3 :

ID	Hypothesis	Criticality
H1	Fleet managers in the target segment experience costly and frequent unplanned downtime and consider it a priority operational problem.	Validates problem–solution fit.
H2	Small-to-mid fleets perceive current predictive-maintenance alternatives as too expensive, complex or OEM-locked.	Validates strategic positioning versus Carly Enterprise, SteerAI and Bosch.
H3	Target fleets can provide sufficiently reliable OBD or telematics data with acceptable integration effort.	Validates data availability and hardware-agnostic positioning.
H4	Maintenance managers consider predictive alerts useful only if they are understandable, prioritized and actionable.	Validates UX and operational adoption.
H5	Fleet managers are willing to pay within the proposed range of USD 8–15 per vehicle per month if value is demonstrated.	Validates revenue model and unit economics.
H6	A pilot deployment can generate measurable improvements in downtime, emergency repairs or maintenance expenditure.	Validates economic impact promised in Section 2.5.

Table 3: Key hypotheses to be validated through market testing.

5.4 Validation Strategy Methods

The validation strategy combines qualitative customer discovery with quantitative pilot evidence, following a staged approach that reduces execution risk before scaling.

5.4.1 Customer Discovery Interviews

We will conduct 15–30 semi-structured interviews with fleet managers and maintenance managers in the target segment. Interviews will explore current maintenance practices with consequent problems, experience with existing tools, data availability and willingness to pay. We state as success criterion that at least 60% of interviewees confirm at least one major obstacle with current solutions (complexity, cost, OEM lock-in or lack of actionable insights), consistent with hypotheses H1 and H2.

5.4.2 Survey

A short online survey (10–12 questions) will be distributed to a broader sample to quantify the prevalence of the problems identified in interviews. The survey will include Likert-scale items on downtime frequency, maintenance cost volatility, satisfaction with current tools and price sensitivity. Results will triangulate interview findings and refine the definition of the target segment.

5.4.3 MVP (Minimum Viable Product)

The MVP will consist of a cloud-based dashboard that ingests OBD or telematics data from a limited set of vehicle models and provides:

1. daily vehicle health scores;
2. ranked alerts with estimated failure probability and recommended actions;
3. a simple cost-savings estimator.

The MVP will not include full fleet optimization or advanced analytics, but only the core predictive-maintenance functionality required to test hypotheses H3, H4 and H6.

5.4.4 Pilot Deployment

We will run a 3–6 month pilot with 2–3 fleet customers (50–100 vehicles in total). The pilot will compare maintenance performance before and after deployment using indicators such as unexpected failures per 10,000 miles, downtime hours per vehicle, emergency repair costs and false-positive rate of alerts. Success criterion: statistically significant improvement in at least two of these metrics (e.g., reduction in breakdowns and downtime) compared to the pre-pilot baseline, supporting hypothesis H6.
All data collected during interviews, surveys and pilots will be anonymized and used solely for research and product development purposes. Customers will be informed about the experimental nature of the MVP and the limitations of the predictive model.

5.5 Learning and Decision-Making

The validation process is designed to generate clear signals and guide iterative development. Table 4 maps each hypothesis to success signals, failure signals and the corresponding decision.

5.5.1 Iterative Development Cycle

After each validation stage (interviews $\rightarrow$ survey $\rightarrow$ MVP $\rightarrow$ pilot), the team will review the evidence and decide whether to: (i) proceed to the next stage; (ii) iterate on the current stage (e.g., refine the model, adjust pricing, improve UX); or (iii) pivot to a different problem, segment or business model. Decisions will be documented and linked to specific data points to ensure transparency and traceability.

Hypothesis	Success signals	Failure signals	Decision
H1 (downtime is a priority problem)	$\geq$60% of interviewees rank unplanned downtime among top 3 operational problems; survey confirms high frequency and cost.	Most interviewees consider downtime manageable or secondary; survey shows low variability in maintenance costs.	Revisit problem definition or target segment; consider pivoting to a different pain point.
H2 (existing solutions are inadequate)	$\geq$60% report at least one major obstacle (complexity, cost, OEM lock-in, lack of actionable insights).	Most respondents are satisfied with current tools or see no clear advantage in a new solution.	Refine positioning or focus on underserved niches (e.g., specific vehicle types or regions).
H3 (data availability)	Pilots achieve stable data ingestion from $\geq$80% of vehicles with acceptable integration effort ($<$2 days per fleet).	Persistent data gaps, incompatible formats or excessive integration costs ($>$1 week per fleet).	Simplify data requirements, partner with telematics providers or narrow the target to fleets with compatible hardware.
H4 (alerts must be actionable)	Maintenance managers report high usability scores; $\geq$70% of alerts are acted upon within 48 hours.	Alerts are ignored or perceived as noisy; low trust in the system.	Improve alert prioritization, explainability and integration with maintenance workflows.
H5 (willingness to pay)	$\geq$50% of interviewees indicate willingness to pay within the proposed range (USD 8–15/vehicle/month) if value is demonstrated.	Most respondents consider the price too high or are unwilling to commit even after seeing pilot results.	Revisit pricing model (e.g., per-fleet flat fee, tiered pricing) or reduce costs to enable lower prices.
H6 (measurable economic impact)	Pilot shows statistically significant reduction in breakdowns, downtime or R&M costs.	No measurable improvement or results are inconsistent across fleets.	Extend pilot duration, refine the model or reconsider the value proposition.

Table 4: Mapping of hypotheses to success signals, failure signals and decisions.

5.5.2 Risk Management

If multiple hypotheses fail simultaneously (e.g., H1, H2 and H5), the venture will be considered not viable in the current form and a strategic pivot will be triggered. If only some hypotheses fail (e.g., H3 or H4), targeted iterations will be prioritized before re-testing.

The validation metrics and decision rules described above provide the basis for the negative backcasting exercise in Section 6. In that analysis, we will assume that the venture has failed and trace backward through these signals to identify which hypotheses (H1–H6) were violated, what early warnings were missed and which mitigation actions could have prevented the failure. This disciplined pessimism complements the validation strategy by highlighting risks that must be monitored even if early tests are successful

6 Failure Analysis andRisk Mitigation

6.1 Scenario of Failure

Assume that by Year 3 the venture has:

- fewer than 5 paying customers and less than 1,000 vehicles under subscription;
- monthly recurring revenue well below the SOM target of USD 2.9 million/year;
- no additional funding and insufficient cash to continue operations;
- at least one pilot customer that discontinued the service after the initial trial.

We now reconstruct how this outcome could have occurred.

6.2 Common Failure Causes

6.2.1 C1 – Weak Problem–Solution Fit (H1 violated)

Fleet managers did not perceive unplanned downtime and reactive maintenance as sufficiently painful or costly. Many considered existing practices (fixed-interval maintenance, ad-hoc repairs) acceptable, especially in smaller fleets with tight budgets. As a result, demand for a predictive-maintenance SaaS tool was lower than expected.

6.2.2 C2 – Inadequate Differentiation versus Existing Solutions (H2 violated)

The venture failed to demonstrate a clear advantage over Carly Enterprise, SteerAI, Bosch or even simple spreadsheet-based approaches. Prospective customers viewed the proposed solution as “another telematics dashboard” without a compelling reason to switch.

6.2.3 C3 – Data Availability and Integration Issues (H3 violated)

Several target fleets lacked reliable OBD or telematics data, or used heterogeneous hardware that was difficult to integrate. Integration efforts took longer and cost more than planned, delaying pilots and increasing onboarding costs.

6.2.4 C4 – Poor Usability and Low Operational Adoption (H4 violated)

Maintenance managers found alerts difficult to interpret, not sufficiently prioritized or not linked to concrete actions. False positives and vague warnings undermined trust in the system, leading to low usage and eventual abandonment.

6.2.5 C5 – Insufficient Willingness to Pay (H5 violated)

Although some managers acknowledged the potential benefits, most were unwilling to pay within the proposed range (USD 8–15/vehicle/month). Price sensitivity was higher than expected, especially among smaller fleets with limited technology budgets.

6.2.6 C6 – Lack of Measurable Economic Impact (H6 violated)

Pilots did not produce statistically significant improvements in breakdowns, downtime or R&M costs. In some cases, results were inconsistent across fleets or vehicle types, making it difficult to generalize the value proposition.

6.2.7 C7 – Strategic and Regulatory Risks

Additional failure paths include: (i) misalignment with broader industry trends (e.g., rapid shift to electric vehicles with different failure modes); (ii) emergence of new regulations on AI in automotive contexts that increase compliance costs; (iii) dependence on third-party telematics providers that change pricing or API access terms.

6.3 Early Warning Signals and Mitigation Actions

For each failure cause, we identify early warning signals that can be monitored during validation and corresponding mitigation actions. Table 5 summarizes this mapping.

Failure cause	**Early warning signals**	**Mitigation actions**
C1 (weak problem–solution fit)	In interviews, downtime ranks outside top 3 problems; low urgency to change current practices.	Refine problem framing; focus on sub-segments with higher downtime costs (e.g., time-critical logistics); consider pivoting to related pain points (e.g., compliance reporting, fuel optimization).
C2 (inadequate differentiation)	Prospects compare the solution unfavorably to existing tools; frequent mentions of "we already have something similar".	Sharpen positioning around specific unmet needs (multi-brand support, interpretable alerts, low integration effort); develop niche features not offered by incumbents.
C3 (data/integration issues)	Pilots require >1 week of integration per fleet; data coverage below 70% of vehicles; repeated format incompatibilities.	Partner with telematics providers for standardized integrations; narrow target to fleets with compatible hardware; invest in data harmonization tools.
C4 (poor usability)	Low alert acceptance rate ($<50\%$); maintenance managers ignore or disable notifications; negative feedback on clarity.	Redesign UX with direct input from maintenance managers; add explainability (e.g., contributing factors, confidence scores); integrate with existing maintenance workflows and ticketing systems.
C5 (low willingness to pay)	Most interviewees reject the proposed price range; requests for free tiers or heavy discounts.	Test alternative pricing models (per-fleet flat fee, tiered pricing, outcome-based contracts); reduce costs to enable lower prices; focus on higher-value segments less price-sensitive.
C6 (no measurable impact)	Pilots show no consistent reduction in breakdowns or costs; high variance across fleets.	Extend pilot duration; refine model features and thresholds; segment by vehicle type and operating conditions; consider hybrid models (predictive + scheduled maintenance).
C7 (strategic/regulatory risks)	New regulations on AI in transportation; key telematics partners change API terms or pricing.	Monitor regulatory developments; diversify data sources; maintain flexibility to adapt the business model (e.g., white-label partnerships).

Table 5: Early warning signals and mitigation actions for each failure cause.

6.4 Strategic Thinking Benefits

The negative backcasting exercise highlights several important insights:

- **Disciplined pessimism.** By assuming failure, the team is forced to confront uncomfortable scenarios that might otherwise be ignored in an optimistic business plan.
- **Prioritization of risks.** The analysis reveals which hypotheses (H1, H5, H6) are most critical to venture viability and should therefore receive the greatest attention during validation.
- **Early detection.** Defining explicit warning signals enables the team to detect problems before they become irreversible and to take corrective action in time.
- **Strategic flexibility.** Identifying multiple failure paths encourages the design of a flexible business model that can pivot or adapt if initial assumptions prove incorrect.

In summary, negative backcasting does not predict that the venture will fail, but rather strengthens the project by making risks visible, measurable and manageable from the outset.